SMALL OFFICE NETWORK DESIGN AND CONFIGURATION

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements of the current semester

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IN

ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by

RAMYA R (192312213)

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

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SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Small Office Network Design and Configuration” has been carried out by RAMYA R (Reg. No. 192312213) and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Electronics and Communication Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

Electronics and Communication Engineering
SIMATS Engineering,
SIMATS, Chennai – 602105.

COURSE FACULTY

Electronics and Communication Engineering
SIMATS Engineering,
SIMATS, Chennai – 602105.

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105

DECLARATION

I, RAMYA R of the Department of Electronics and Communication Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Small Office Network Design and Configuration” is the result of my own bonafide efforts. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated.

Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. Cisco Packet Tracer was used as the network simulation environment. An AI writing assistant (Claude, developed by Anthropic) was used to help draft and format this report from the project details, configuration, test observations and screenshot supplied by me; the results reported are those observed in my Packet Tracer simulation. I confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Signature of the Student with Name

RAMYA R (192312213)

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

Individual Contribution Statement

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
RAMYA R (192312213)	Sole student; all project tasks	Topology design, IPv4 addressing plan, router and server configuration in Cisco Packet Tracer	Ping connectivity tests, DHCP configuration attempt and troubleshooting, results analysis	All chapters, tables, figures and appendices	100%
Total					**100%**

This project was carried out by a single student; the contribution is therefore 100% as shown above.

ABSTRACT

Small offices depend on a reliable local area network (LAN) to share files, services and a single gateway, yet network configuration is often learned as isolated commands rather than as a complete, tested design. This capstone project addresses the engineering problem of designing, configuring and verifying a small office LAN with one router, one Ethernet switch, four client computers and one server, using a consistent IPv4 addressing plan and without access to physical equipment.

The work is significant because it demonstrates, at no hardware cost and with no equipment risk, the building blocks on which every small office network relies: Ethernet switching, IPv4 addressing, default-gateway configuration and connectivity testing.

The proposed solution is a star-topology LAN simulated in Cisco Packet Tracer. A Cisco 1941 router (Router0) connects to a Cisco 2960-24TT switch (Switch0), to which PC0, PC1, PC2, PC3 and Server0 are attached with copper straight-through cables. All devices share the private network 192.168.1.0/24. Router0 uses 192.168.1.1 on GigabitEthernet0/0 as the default gateway, Server0 uses 192.168.1.2, and the PCs use the static addresses 192.168.1.10 to 192.168.1.13.

The methodology comprised requirement definition, comparison of three alternative designs with a weighted matrix, construction of the topology, configuration of the router and hosts, a DHCP configuration attempt on Server0, and connectivity testing with ping.

All cable links showed green link indicators. PC0 pinged Router0 successfully with four replies and 0% packet loss, and successful replies were also reported from PC0 to PC1, PC2 and PC3. A DHCP pool named OfficePool was configured on Server0, but during troubleshooting a client displayed a default gateway of 0.0.0.0, so DHCP gateway assignment was not verified. DNS name resolution, internet connectivity, VLANs, security features, throughput and physical hardware performance were not tested.

The principal conclusion is that the statically addressed design meets the core connectivity objectives in simulation, while DHCP operation and security hardening remain open items for future work.

Keywords: Small office network; Local area network; IPv4 addressing; Ethernet switching; Default gateway; Cisco Packet Tracer

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
ACL	Access control list	—
CIDR	Classless Inter-Domain Routing	—
DHCP	Dynamic Host Configuration Protocol	—
DNS	Domain Name System	—
Gi	GigabitEthernet (interface name prefix)	—
ICMP	Internet Control Message Protocol	—
IEEE	Institute of Electrical and Electronics Engineers	—
IETF	Internet Engineering Task Force	—
IP	Internet Protocol	—
IPv4	Internet Protocol version 4	—
LAN	Local Area Network	—
MAC	Media Access Control	—
PC	Personal computer	—
PC-PT / Server-PT	Packet Tracer end-device and server models	—
RFC	Request for Comments	—
SO	Student Outcome	—
VLAN	Virtual Local Area Network	—
/24	Network prefix length (mask 255.255.255.0)	bits

CHAPTER 1
INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

A computer network allows two or more devices to exchange data over a shared medium using agreed rules known as protocols. A local area network (LAN) is a network that covers a limited area such as a single office floor. In a small office, the LAN typically connects a handful of desktop computers and one or more shared servers, and it provides a single path, through a router, towards other networks. Ethernet is the dominant wired LAN technology and is standardised in IEEE 802.3 [8].

Two devices form the core of such a LAN. An Ethernet switch connects the end devices and forwards each frame only towards the port where its destination was learned, while a router joins the LAN to other networks and acts as the default gateway for any traffic that must leave the local subnet. Each device is identified by an IPv4 address [1]. Private address blocks reserved for internal use, such as 192.168.0.0/16, are defined in RFC 1918 [3], so a small office can address its devices without registering public addresses.

Designing even a small network requires decisions about topology, addressing, services and verification. Because physical routers and switches are costly and require laboratory access, network simulators such as Cisco Packet Tracer [11] are widely used to design and test configurations before, or instead of, physical deployment. This project applies that approach to a small office LAN consisting of a Cisco 1941 router, a Cisco 2960-24TT switch, four PCs and one server.

1.2 Need for the Project

Why the problem needs to be addressed. A small office network that is configured without a clear addressing plan or without systematic testing is prone to duplicate addresses, wrong default gateways and connectivity faults that are time-consuming to diagnose. A documented, tested reference design reduces these faults.

Who is affected. Office staff depend on the network to reach shared resources; the person who administers the network (often a non-specialist) must be able to understand and maintain it; and students learning networking need a complete worked example rather than isolated commands.

Existing limitations. Many introductory exercises stop at configuring a single device or show only a successful ping, without recording which features were verified and which were not.

Service integration, such as DHCP on a server, is frequently assumed to work once configured, although a misconfigured pool can leave clients with incomplete settings.

Engineering significance. The project integrates topology design, IPv4 subnetting, device configuration, service configuration and verification in one traceable design, and it records honestly where verification succeeded and where it did not. This practice of separating verified from unverified functionality is central to sound engineering reporting.

1.3 Problem Statement

To design, configure and verify, in a simulated environment, a small office LAN that connects four client computers and one server through an Ethernet switch to a router acting as default gateway, using a single IPv4 subnet, so that all devices can communicate reliably, while documenting which functions (static addressing, DHCP, DNS) are verified and which are not.

The problem involves several interacting factors and conflicting requirements:

- **Interacting factors:** physical connectivity, IPv4 address and mask consistency across seven configured interfaces, a correct default gateway on every host, and the interaction between a server-hosted DHCP service and statically addressed hosts.
- **Conflicting requirements:** a flat single subnet is simple and cheap but offers no segmentation or access control; automatic addressing with DHCP reduces administration but adds a service that must itself be configured correctly; a simulation removes cost and hardware risk but cannot demonstrate physical performance.
- **Realistic constraints:** no physical hardware, no hardware budget, and a defined set of devices (one router, one switch, four PCs, one server) that may not be extended.

1.4 Project Aim

The aim of this project is to design and simulate a small office Local Area Network in Cisco Packet Tracer that demonstrates correct network topology, IPv4 addressing, Ethernet switching, router configuration, server integration and connectivity testing.

1.5 Project Objectives

1. To design a star-topology LAN of one router, one switch, four PCs and one server with an IPv4 addressing plan for the subnet 192.168.1.0/24 in which no address is duplicated.
2. To construct the design in Cisco Packet Tracer using the specified devices and copper straight-through cables, with every link showing a green link indicator.

3. To configure GigabitEthernet0/0 of Router0 with 192.168.1.1/24 and enable it as the default gateway of the LAN.
4. To assign static IPv4 settings (address, mask, default gateway and DNS server) to Server0 and to PC0, PC1, PC2 and PC3.
5. To test connectivity by pinging Router0 from PC0 (target: four replies, 0% loss) and by pinging PC1, PC2 and PC3 from PC0.
6. To configure a DHCP pool named OfficePool on Server0 and assess whether a client receives the correct default gateway.
7. To evaluate the results against the objectives and to state the limitations of the work honestly.

1.6 Scope

Table 1.1 summarises what is included in and excluded from the project, together with the assumptions and boundary conditions.

Table 1.1: Scope, assumptions and boundary conditions

Category	Details
Included	Topology design; IPv4 plan for 192.168.1.0/24; Router0 interface configuration; static addressing of Server0 and PC0–PC3; ping tests; DHCP pool configuration attempt on Server0; analysis and documentation.
Excluded	Internet or WAN connectivity; VLANs; firewall rules or access control lists; wireless access; DNS name-resolution testing; security testing; throughput testing; deployment on physical hardware.
Assumptions	The Packet Tracer devices behave like their physical counterparts for basic switching, addressing and ICMP; all devices belong to one organisation and one broadcast domain.
Boundary conditions	Single subnet 192.168.1.0/24; one router interface in use; six cables to Switch0; simulation only.

1.7 Expected Engineering Outcome

The expected outcome is a working simulated infrastructure solution: a documented small office LAN prototype in Cisco Packet Tracer with a complete addressing plan, a saved router configuration, statically addressed hosts, and a test record that distinguishes verified connectivity from features that remain unverified. The prototype can serve as a baseline for later extension, for example with VLAN segmentation, access control and physical deployment.

CHAPTER 2
LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review covered technical standards, vendor documentation and established networking practice rather than research papers. The Internet Engineering Task Force (IETF) Request for Comments (RFC) documents were consulted for IPv4, ICMP, private addressing, CIDR notation, DHCP and DNS [1]–[7]; the IEEE 802.3 Ethernet standard was consulted for wired LAN technology [8]; and Cisco product documentation and the Cisco Networking Academy description of Packet Tracer were consulted for the devices and the simulation tool [9]–[11]. No academic papers, patents or performance statistics are cited, and none are claimed.

2.2 Review of Existing Work

2.2.1 Small-office network designs

The usual small-office design is a star topology: end devices connect by twisted-pair copper to a central switch, and the switch connects to a router that provides the default gateway. All devices normally share one IP subnet. This design is easy to understand and inexpensive, and it is the baseline adopted in this project. Larger or more security-sensitive offices extend it with VLANs, access control lists, redundant links and wireless access points.

2.2.2 Ethernet switches

An Ethernet switch operates at the data-link layer. It learns the source MAC address of each frame it receives and then forwards later frames only to the port where the destination address was learned, which avoids the collisions and unnecessary traffic of shared-medium hubs. Ethernet frame formats and the copper physical layers are defined in IEEE 802.3 [8]. The Cisco Catalyst 2960 family is a fixed-configuration Ethernet switch line aimed at entry-level, mid-market and branch-office networks [10]; the 2960-24TT model is used in this project.

2.2.3 Router and default-gateway configuration

A router forwards packets between IP networks. Hosts send any packet whose destination lies outside their own subnet to the default gateway, which is the router interface on the local subnet. A router interface must have an IPv4 address and mask and must be administratively enabled before it forwards traffic. The Cisco 1941 is a member of the Cisco 1900 Series Integrated Services Routers, which target small businesses and branch offices [9].

2.2.4 IPv4 addressing and subnet masks

An IPv4 address is a 32-bit number written as four decimal octets [1]. The subnet mask identifies which bits belong to the network prefix; 255.255.255.0 corresponds to a /24 prefix in CIDR notation [7]. A /24 network such as 192.168.1.0/24 contains $2^8 = 256$ addresses, of which 254 can be assigned to hosts because the first address (192.168.1.0) identifies the network and the last (192.168.1.255) is the broadcast address. The block 192.168.0.0/16 is reserved for private use [3], so 192.168.1.0/24 is suitable for an internal office network.

2.2.5 Static addressing and DHCP

With static addressing, an administrator enters the address, mask, gateway and DNS server on each device. This is predictable and needs no extra service, but it does not scale and is prone to typing errors. The Dynamic Host Configuration Protocol (DHCP) [4] lets a server lease these parameters to clients automatically through a four-step exchange (discover, offer, request, acknowledge). The server is configured with an address pool and options such as the default gateway and DNS server. Infrastructure devices such as routers and servers are normally addressed statically, outside the DHCP pool, so that their addresses never change.

2.2.6 DNS services

The Domain Name System (DNS) [5], [6] maps human-readable names to IP addresses. In a small office, a local server can answer name queries for internal hosts and forward other queries. In this project Server0 is intended to hold the DNS role, and 192.168.1.2 is handed to clients as the DNS server address, but name resolution was not tested.

2.2.7 Network simulation and Cisco Packet Tracer

Cisco Packet Tracer is a network simulation and visualisation tool from the Cisco Networking Academy that allows learners to build topologies and configure virtual devices without hardware [11]. It supports command-line configuration of virtual routers and switches and graphical configuration of end devices and servers. Its limits are that device behaviour is simulated, so timing, throughput and failure behaviour of real equipment are not reproduced.

2.3 Comparative Analysis

Table 2.1: Comparison of existing approaches

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Flat LAN with one switch and one router, static addressing	Simple, low cost, predictable addresses	No segmentation; manual configuration does not scale	Basis of the proposed design
Flat LAN with DHCP on a server or router	Automatic configuration of clients; fewer typing errors	Adds a service that must be configured and verified	DHCP attempted on Server0; gateway assignment not verified
VLAN-segmented LAN (router-on-a-stick)	Traffic separation; improved security posture	More configuration; needs trunking and sub-interfaces	Evaluated as Alternative 2; excluded from scope
Wireless LAN with access point	No cabling to clients; flexible placement	Radio security and interference concerns	Evaluated as Alternative 3; not adopted
Physical hardware laboratory	Real performance and fault behaviour	Equipment cost, space and access required	Not available; simulation used instead

2.4 Research / Engineering Gap

Standards and vendor documents describe each technology separately, and introductory exercises usually show only a successful configuration. What remains unresolved for a learner is a compact, fully documented design that ties the topology, addressing plan, configuration and test evidence together and that states explicitly which functions were verified, which were configured but unverified, and which were not tested. This is the gap addressed by the project.

2.5 Proposed Contribution

The project contributes a complete, traceable design of a small office LAN in Cisco Packet Tracer. It differs from a typical configuration exercise in four ways: it compares alternative designs with a weighted matrix before selecting one; it keeps one consistent addressing plan across the whole report; it records a test plan before the results; and it classifies each feature as passed, not verified or not tested. In particular, it reports the unresolved DHCP gateway observation rather than omitting it.

CHAPTER 3
ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Table 3.1: Stakeholder requirements

Stakeholder	Requirement
Office staff (PC users)	Reliable connection between their PCs, the server and the gateway; no manual network settings in the long term
Network administrator	A simple, documented design with a clear addressing plan and a saved router configuration
Student / designer	A design that can be built and tested entirely in simulation with the specified devices
Course faculty / assessor	Evidence of design, testing and honest reporting of results and limitations

Functional & Non-Functional Requirements

Table 3.2: Functional and non-functional requirements

Requirement	Type (Functional / Non-Functional)	Measurable Target
FR1: All devices share one LAN	Functional	Six devices connected to Switch0 within 192.168.1.0/24
FR2: Router acts as default gateway	Functional	Router0 Gi0/0 = 192.168.1.1/24, interface enabled
FR3: Hosts have valid static settings	Functional	Server0 and PC0–PC3 configured with address, mask, gateway 192.168.1.1 and DNS 192.168.1.2
FR4: Connectivity is verifiable	Functional	PC0 to Router0: 4 replies, 0% loss; PC0 to PC1, PC2, PC3: replies received
FR5: Server hosts DHCP role	Functional	Pool OfficePool: start 192.168.1.100, 50 users; client receives gateway 192.168.1.1
NFR1: Simplicity	Non-Functional	One subnet, one router interface in use, one switch
NFR2: Growth headroom	Non-Functional	18 of 24 switch access ports unused (24 minus 6 cables)
NFR3: Low cost	Non-Functional	No hardware purchase for the simulation
NFR4: Maintainability	Non-Functional	Configuration saved with write memory; addressing documented in one table
NFR5: Security awareness	Non-Functional	Security gaps identified and recorded in the risk register (Section 3.7)

3.2 Constraints & Applicable Standards

The constraints that apply to this project are technical (simulation only, a fixed device list, one subnet), economic (no hardware budget) and ethical (results must be reported as observed). Environmental, regulatory and health-and-safety constraints do not materially restrict a software simulation; they are discussed for a physical deployment in Section 3.7. The specific standards applied are listed in Table 3.3.

Table 3.3: Applicable standards and how they were applied

Standard / Code	Requirement	How Applied in This Project
IEEE Std 802.3 (Ethernet) [8]	Defines Ethernet frames and twisted-pair copper physical layers	Copper straight-through Ethernet cables between router, switch, PCs and server
RFC 791 (IPv4) [1]	IPv4 addressing and packet format	Single IPv4 plan; every device has a unique address
RFC 792 (ICMP) [2]	Echo request and echo reply messages	Ping used as the connectivity test
RFC 1918 [3]	Private address blocks for internal networks	Subnet 192.168.1.0/24 lies in 192.168.0.0/16
RFC 4632 (CIDR) [7]	Prefix-length notation	Network written as 192.168.1.0/24 (mask 255.255.255.0)
RFC 2131 (DHCP) [4]	Automatic lease of address and options	DHCP pool OfficePool configured on Server0 (gateway assignment not verified)
RFC 1034 / 1035 (DNS) [5], [6]	Domain name resolution	Server0 intended as DNS server (192.168.1.2); resolution not tested

3.3 Design Specifications

Table 3.4: Design specifications

Parameter	Target / Requirement	Unit	Priority
Network address	192.168.1.0/24 (mask 255.255.255.0)	IPv4 prefix	High
Usable host addresses	254	addresses	Medium
Router LAN interface	GigabitEthernet0/0 = 192.168.1.1	IPv4 address	High
Statically addressed hosts	5 (Server0, PC0, PC1, PC2, PC3)	devices	High
Devices cabled to Switch0	6 (Router0, four PCs, Server0)	cables	High
Cable type	Copper straight-through Ethernet	—	High
DHCP pool (intended)	Start 192.168.1.100, maximum 50 users	addresses	Medium
Packet loss, PC0 to Router0	0	%	High
Hardware cost of simulation	None required	—	Medium

Table 3.5: Device descriptions

Device (Name)	Model / Type	Role in the Network
Router0	Cisco 1941 router [9]	Default gateway for the LAN; interface Gi0/0 faces Switch0
Switch0	Cisco 2960-24TT switch [10]	Central Ethernet switch connecting all devices
PC0, PC1, PC2, PC3	PC-PT end devices	Office client computers with static addresses
Server0	Server-PT	Server with static address; intended DHCP and DNS role
Cables	Copper straight-through Ethernet	Six links, all to Switch0

Table 3.6: IPv4 addressing plan

Device	IPv4 Address	Subnet Mask	Default Gateway	DNS Server
Router0 GigabitEthernet0/0	192.168.1.1	255.255.255.0	Not applicable	Not applicable
Server0	192.168.1.2	255.255.255.0	192.168.1.1	192.168.1.2
PC0	192.168.1.10	255.255.255.0	192.168.1.1	192.168.1.2
PC1	192.168.1.11	255.255.255.0	192.168.1.1	192.168.1.2
PC2	192.168.1.12	255.255.255.0	192.168.1.1	192.168.1.2
PC3	192.168.1.13	255.255.255.0	192.168.1.1	192.168.1.2

Network address: 192.168.1.0/24. Subnet mask: 255.255.255.0. Default gateway: 192.168.1.1.

3.4 Alternative Solutions & Evaluation

Alternative 1 – Flat LAN with one switch and one router. A single Cisco 2960-24TT switch connects the router, four PCs and the server in one subnet, 192.168.1.0/24, with the router as default gateway. It is the simplest design and uses only the specified devices.

Alternative 2 – VLAN-segmented LAN (router-on-a-stick). The same devices are placed in separate VLANs (for example staff and server), joined through a trunk and router sub-interfaces. It limits broadcast and access between groups but needs trunking and sub-interface configuration beyond the scope of this project.

Alternative 3 – Wireless LAN with an access point. Clients connect by Wi-Fi to an access point attached to the router. It avoids client cabling but needs radio security configuration and is not part of the specified device list.

The criteria were weighted according to the project priorities (connectivity and reliability highest, then cost). Each alternative was scored from 1 (poor) to 5 (best) by the author's qualitative judgement; Safety includes exposure to unauthorised access.

Table 3.7: Weighted evaluation matrix (scores 1–5, weighted total out of 5)

Criterion	Weight (%)	Alternative 1	Alternative 2	Alternative 3
Performance	25	4	4	3
Cost	20	5	3	3
Safety	15	3	5	2
Sustainability	15	4	3	3
Reliability	25	4	3	3
Weighted total	**100**	**4.05**	**3.55**	**2.85**

3.5 Selected Design & Detailed Design

Alternative 1 was selected. It has the highest weighted total (4.05 against 3.55 and 2.85, Table 3.7), it meets every objective in Section 1.5 using only the specified devices, and it avoids the extra configuration and device cost of the alternatives. Alternative 2 scores higher on safety, but its additional complexity is outside the scope of this project; it is retained as future work.

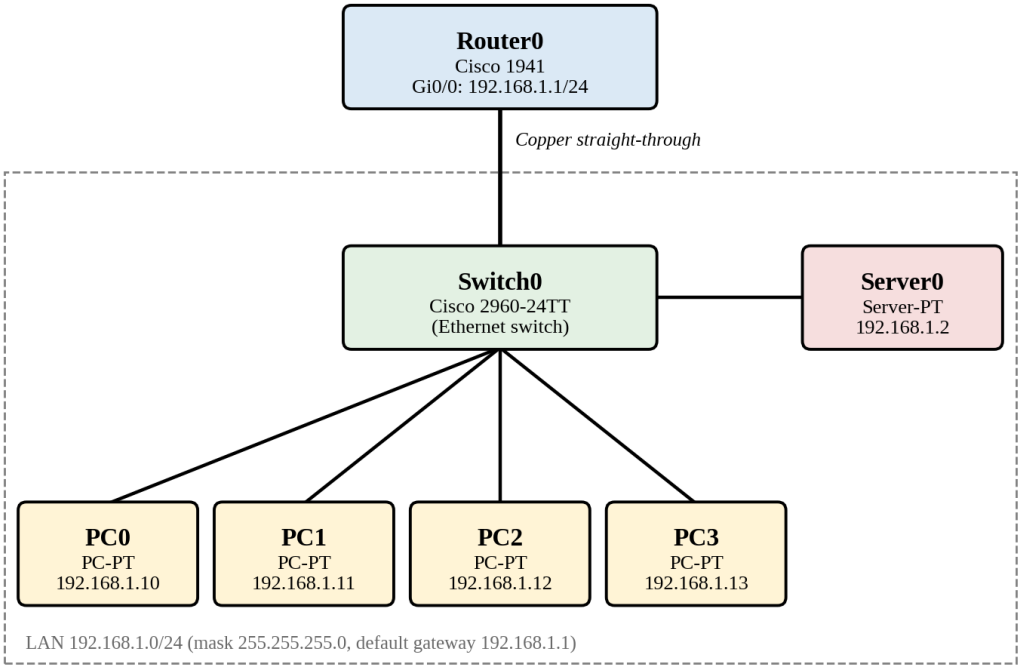


Figure 3.1: Network Architecture of the Small Office LAN (Router0 – Switch0 – PC0, PC1, PC2, PC3, Server0)

Figure 3.1 shows the architecture. Router0 connects only to Switch0; PC0, PC1, PC2, PC3 and Server0 each connect directly to Switch0. No end device is connected directly to the router. The layout matches the Packet Tracer screenshot in Figure 4.1.

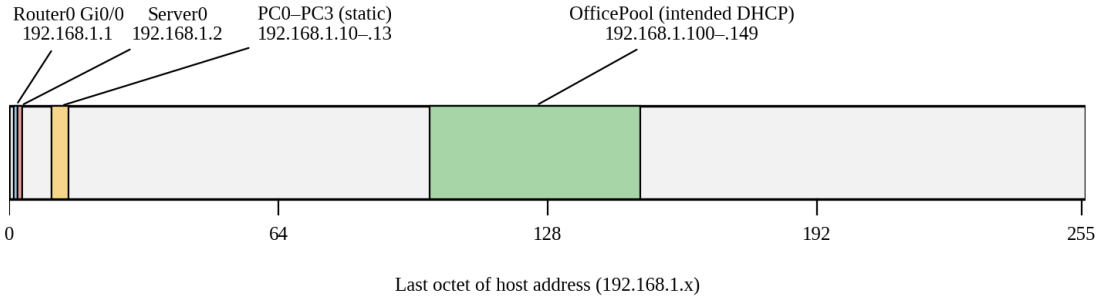


Figure 3.2: IPv4 Address Allocation within 192.168.1.0/24

Key calculations. The essential address calculations are summarised in Table 3.8; the derivation is given in Appendix A.

Table 3.8: Address-space summary for 192.168.1.0/24

Quantity	Value
Usable host addresses ($2^8 - 2$)	254
Statically assigned addresses (Router0, Server0, PC0–PC3)	6
Intended DHCP pool (192.168.1.100 to 192.168.1.149)	50
Addresses neither assigned nor in the pool	254 − 6 − 50 = 198

Systems approach. Switch0 is the common interface: every device reaches every other device through it, so its operation is a dependency for all traffic. Router0 is needed only for traffic leaving the subnet and for the default-gateway entry on each host. Server0 depends on Switch0 for reachability and is intended to supply DHCP and DNS parameters to clients, which in turn depend on Router0’s address as their gateway. Keeping the server and router addresses outside the DHCP pool (.1 and .2 against a pool starting at .100) prevents the service from leasing an address already in use.

3.6 Trade-off Analysis

Table 3.9: Design trade-offs

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Single flat subnet	VLAN segmentation	Simplicity, lowest configuration effort	No traffic separation or access control	Flat subnet; highest weighted score and within scope
Static addressing used for connectivity tests	DHCP for all clients	Predictable addresses; tests not dependent on DHCP	Manual entry; does not scale	Static; DHCP gateway assignment was not verified
Server-based DHCP and DNS role	DHCP provided by the router	Uses the specified server; separates services from routing	Server becomes a service dependency	Server role attempted; DHCP outcome unverified
One 24-port switch	Several smaller switches	One device to manage; growth headroom	Single point of failure	One switch; redundancy left to future work
/24 prefix	Smaller prefix (for example /26)	Room for growth; easy to read	Larger broadcast domain than needed for six devices	/24; conventional and simple for a small office

3.7 Sustainability, Ethics, Safety, Risk & Security

Table 3.10: Sustainability, ethics, health and safety, and security considerations

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Equipment manufacture, energy use and e-waste	Budget (Table 4.13): no hardware purchased for the simulation	Design and test performed in simulation; physical purchase deferred until a deployment is justified
Sustainability	Over-provisioning and early replacement	Topology: 6 of 24 switch access ports used	One 24-port switch kept, so growth needs no new switch
Ethics	Honest reporting of results	DHCP client showed gateway 0.0.0.0 during troubleshooting	DHCP reported as NOT VERIFIED, not as working
Ethics	Privacy and professional conduct	Only simulated devices and private addresses; no real user data	No personal data used; sources and AI assistance acknowledged
Health & Safety	Electrical and cabling hazards (physical deployment only)	Simulation involves no mains-powered equipment	No safety controls needed now; cable routing and equipment ventilation noted for deployment
Health & Safety	Ergonomic workstation layout	Not applicable to a simulation	Not evaluated
Security	Unrestricted access within one flat subnet	Topology: all devices in one broadcast domain; no VLANs or ACLs	Recorded as risk R4; VLANs and ACLs listed as future work
Security	Router management protection	Router configuration (Section 4.2.4) contains no password or remote-access settings	Recorded as risk R4; excluded from scope and listed as future work
Security	Default gateway integrity	Gateway 0.0.0.0 observed on a DHCP client	Static gateway used for tests; DHCP to be validated before reliance

Risk Register

Table 3.11: Risk register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
R1: DHCP clients receive an incorrect default gateway	High (observed)	Medium	High	Use static addressing; validate pool settings before reliance	Medium
R2: Duplicate IP address between static hosts and DHCP leases	Low	Medium	Low–Medium	Static hosts at .1–.13; pool starts at .100	Low
R3: Single point of failure (Switch0 or Router0)	Medium	High	High	Document recovery; add redundant devices in a real deployment	Medium
R4: Unauthorised access (flat subnet, no passwords or ACLs)	Medium	High	High	Excluded from scope; add VLANs, ACLs and device passwords	Medium–High
R5: Simulation does not reflect physical behaviour	Medium	Medium	Medium	State clearly that hardware was not tested; plan hardware validation	Medium
R6: Router configuration lost after restart	Low	Medium	Low–Medium	write memory saved the configuration	Low

CHAPTER 4
IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project followed an incremental approach. The topology was built first and its links checked, then the router interface and the static host settings were configured and verified with ping, and finally a DHCP service was configured on Server0 and its effect on a client was investigated. Each stage was tested before the next was added, so that a fault could be traced to the most recent change. The resources used are listed in Table 4.1.

Table 4.1: Resources used

Resource	Description
Simulation environment	Cisco Packet Tracer
Devices	Router0 (Cisco 1941), Switch0 (Cisco 2960-24TT), PC0–PC3 (PC-PT), Server0 (Server-PT)
Cabling	Copper straight-through Ethernet cables (simulated)
Evidence	Packet Tracer topology screenshot and recorded test observations

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

4.2.1 Software and simulation environment

The network was implemented entirely in software in Cisco Packet Tracer [11]. It was not deployed on a physical network. No physical devices were purchased, wired or powered.

4.2.2 Device selection and network construction

One Cisco 1941 router (Router0), one Cisco 2960-24TT switch (Switch0), four PC-PT computers (PC0, PC1, PC2, PC3) and one Server-PT (Server0) were placed in the workspace. These are the only devices in the project. The router was placed above the switch and the end devices below and beside it so that the star topology is clear.

4.2.3 Cable connections

Six copper straight-through Ethernet cables were used, all terminating on Switch0, as listed in Table 4.2. The screenshot (Figure 4.1) does not show port names, so port numbers are not stated.

Table 4.2: Cable connections

Link	Device A	Device B	Cable Type
1	Router0 (GigabitEthernet0/0)	Switch0	Copper straight-through
2	PC0	Switch0	Copper straight-through
3	PC1	Switch0	Copper straight-through
4	PC2	Switch0	Copper straight-through
5	PC3	Switch0	Copper straight-through
6	Server0	Switch0	Copper straight-through

After the cables were connected, the link indicators at both ends of every cable showed green in Packet Tracer, as visible in Figure 4.1.

4.2.4 Router configuration

The following commands were entered on Router0 to configure the LAN interface:

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
end
write memory
```

The purpose of each command is explained in Table 4.3.

Table 4.3: Purpose of the router commands

Command	Purpose
enable	Enters privileged EXEC mode, which is needed before configuration commands can be used
configure terminal	Enters global configuration mode
interface gigabitEthernet 0/0	Selects the router interface that connects to Switch0
ip address 192.168.1.1 255.255.255.0	Assigns the IPv4 address and subnet mask; this address is the default gateway of the LAN
no shutdown	Enables the interface; router interfaces are administratively down until this is entered
exit	Leaves interface configuration mode
end	Returns to privileged EXEC mode
write memory	Saves the running configuration to the startup configuration so it is kept after a restart

4.2.5 IPv4 configuration

Static IPv4 settings were assigned to every device as given in the addressing plan (Table 3.6). Router0 uses 192.168.1.1 on GigabitEthernet0/0; Server0 uses 192.168.1.2; PC0, PC1, PC2 and PC3 use 192.168.1.10, 192.168.1.11, 192.168.1.12 and 192.168.1.13. All use the mask 255.255.255.0, and the server and PCs use the default gateway 192.168.1.1 and the DNS server 192.168.1.2. Static IPv4 addressing was used for all of the successful connectivity tests.

4.2.6 Server configuration

Server0 was assigned the IP address 192.168.1.2, subnet mask 255.255.255.0 and default gateway 192.168.1.1. Its intended role is to provide DHCP and DNS services. Only the DHCP configuration was attempted; DNS resolution was not tested.

4.2.7 DHCP configuration attempt and troubleshooting

A DHCP pool named OfficePool was configured on Server0 with the settings in Table 4.4. These are the intended settings of the attempted configuration.

Table 4.4: Intended DHCP pool settings on Server0

Parameter	Value
Pool name	OfficePool
Default gateway	192.168.1.1
DNS server	192.168.1.2
Start IP address	192.168.1.100
Subnet mask	255.255.255.0
Maximum number of users	50

During troubleshooting, a client displayed a default gateway of 0.0.0.0. The DHCP gateway assignment was therefore **not verified as working correctly**. The cause was not isolated. Possible causes, none of which was confirmed, include a pool setting that was not saved or applied, the service not being active, or the client retaining earlier settings. The issue remains unresolved; static addressing was used to continue the connectivity tests.

Table 4.5: Tools and equipment used

Tool / Software / Equipment	Purpose	Stage Used
Cisco Packet Tracer	Build and simulate the network	Topology construction and all testing
Router command-line interface	Configure and save the Gi0/0 interface	Router configuration
Host IP settings (PCs and Server0)	Enter static address, mask, gateway and DNS	IPv4 configuration
Server0 DHCP service settings	Configure pool OfficePool	DHCP configuration attempt
Ping utility on PC0	Test reachability using ICMP echo [2]	Connectivity testing
Screenshot capture	Record topology evidence	Documentation

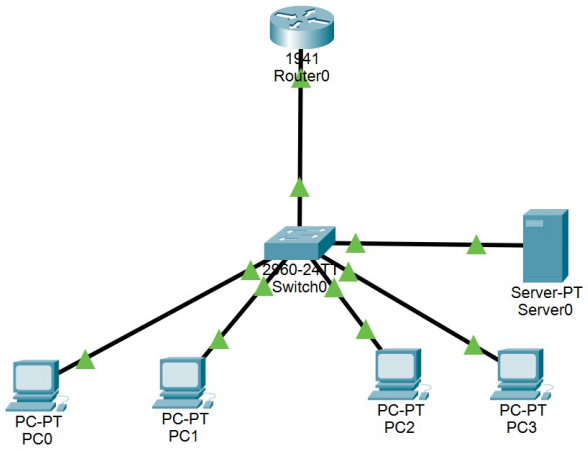


Figure 4.1: Small Office Network Topology Implemented in Cisco Packet Tracer

Figure 4.1 shows Router0 (1941) connected to Switch0 (2960-24TT), with PC0, PC1, PC2, PC3 and Server0 each connected to Switch0. Green indicators are visible on the links. The screenshot does not show port names or IP addresses, so those details are taken from the configuration records in this report and not from the image.

4.3 Test Plan & Verification

The test plan in Table 4.6 was defined before the results were collected. The acceptance criteria apply only to the features listed.

Table 4.6: Test plan

Requirement	Test Method	Acceptance Criterion
T1: Physical links are up	Inspect link indicators in the Packet Tracer topology	Every cable shows green link indicators
T2: Router is reachable from a host	Ping 192.168.1.1 from PC0	Four replies, 0% packet loss
T3: Hosts reach each other through Switch0	Ping PC1, PC2 and PC3 from PC0	Successful replies received
T4: Server uses the planned static settings	Review Server0 IP settings	192.168.1.2 / 255.255.255.0 / gateway 192.168.1.1
T5: DHCP supplies the correct gateway	Configure OfficePool; inspect a DHCP client's gateway	Client shows default gateway 192.168.1.1
T6: Physical hardware performance	Would require physical devices	Outside the scope of this simulation

Table 4.7: Specification compliance summary

Specification	Required Value	Achieved Value	Status
Link status (T1)	All links green	Green link indicators observed	PASS (observed)
PC0 to Router0 (T2)	4 replies, 0% loss	4 replies, 0% loss	PASS
PC0 to PC1, PC2, PC3 (T3)	Successful replies	Successful replies reported	PASS (as reported)
Server0 static settings (T4)	192.168.1.2 / 255.255.255.0 / 192.168.1.1	Assigned as specified	Configured
DHCP gateway assignment (T5)	Client gateway 192.168.1.1	Client displayed 0.0.0.0	NOT VERIFIED
Physical hardware performance (T6)	Not applicable to simulation	Not tested	NOT TESTED

4.4 Results

Table 4.8 gives the connectivity results. Only results provided from the implementation are reported. The ping output itself is not reproduced; the counts are those observed.

Table 4.8: Connectivity test results

Test	Source → Destination	Addressing	Observed Result	Status
T2	PC0 → Router0 (192.168.1.1)	Static	Four replies, 0% packet loss	PASS
T3a	PC0 → PC1 (192.168.1.11)	Static	Successful replies reported	PASS (as reported)
T3b	PC0 → PC2 (192.168.1.12)	Static	Successful replies reported	PASS (as reported)
T3c	PC0 → PC3 (192.168.1.13)	Static	Successful replies reported	PASS (as reported)

Table 4.9 classifies the feature areas, including those that were not verified or not tested.

Table 4.9: Classification of features

Feature	Classification
Router connectivity	PASS
PC-to-PC connectivity	PASS as reported
DHCP gateway assignment	NOT VERIFIED
Physical hardware performance	NOT TESTED
DNS name resolution; internet connectivity; VLANs; firewall rules; security testing; throughput	NOT TESTED (no claim made)

4.5 Data Analysis & Engineering Judgment

The successful ping from PC0 to Router0 with no packet loss shows that the physical link, the switch, the matching address and mask on both devices, and the enabled router interface all work together on the 192.168.1.0/24 subnet. The successful replies from PC0 to PC1, PC2 and PC3 show that Switch0 forwards traffic between the PCs. Because the statically assigned addresses were unique and in one subnet, no address-related fault appeared.

The results have limits. The PC-to-PC tests were reported from PC0 only, each test was run once, and ICMP success does not prove that application services work. Server0 reachability and DNS were not reported as tested. Because static addressing was used, the successful tests say nothing about DHCP.

The DHCP observation deviated from expectation: a client showing gateway 0.0.0.0 indicates that the pool’s gateway option was not delivered or applied. The engineering implication is practical. A client with a valid address and mask but no gateway can still talk to other hosts on its own subnet, so local pings may succeed, but it cannot reach any other network. The fault

would therefore remain hidden in local tests and appear only when off-subnet access is added. For this reason DHCP must not be relied on until the gateway is shown to be delivered correctly, and static addressing is the safer choice for the router and server in any case.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Compared with the baseline flat small-office design in Section 2.3, the project matches its basic switching, addressing and gateway functions, and it adds a documented design process and a clear record of what was and was not verified. It does not match the segmentation of a VLAN design, the realism of a hardware laboratory, or the automatic configuration of a working DHCP deployment. Table 4.10 compares the outcomes with the objectives of Section 1.5.

Table 4.10: Achievement of objectives

Objective	Evidence	Achievement (Fully / Partially / Not Achieved)
1. Design the LAN and addressing plan	Figure 3.1; Tables 3.6 and 3.8	Fully
2. Construct the design in Packet Tracer	Figure 4.1; green link indicators; Table 4.2	Fully
3. Configure Router0 Gi0/0 as gateway	Section 4.2.4; PC0 → Router0 ping, 0% loss	Fully
4. Assign static settings to Server0 and PCs	Table 3.6; Section 4.2.5	Fully
5. Test connectivity by ping	Table 4.8	Fully (for the tests listed)
6. Configure DHCP and assess gateway assignment	Table 4.4; client gateway 0.0.0.0	Partially (configured; gateway assignment not verified)
7. Evaluate results and state limitations	Sections 4.5 and 4.7	Fully

4.7 Strengths & Limitations

Strengths:

- Simple, easily understood star topology that uses only the specified devices.
- Consistent addressing plan with no duplicate addresses and clear separation of static addresses from the intended DHCP pool.
- Router connectivity verified with 0% packet loss, and PC-to-PC connectivity reported as successful.
- Test plan defined before results, with unverified and untested features stated openly.
- No hardware cost, and the configuration was saved on the router.

Limitations:

- Implemented only in simulation; physical hardware performance was not tested.
- DHCP gateway assignment is not verified; a client displayed gateway 0.0.0.0 and the cause is unresolved.
- DNS name resolution, internet connectivity, VLANs, firewall rules, security and throughput were not tested.
- Single flat subnet with no access control, and single points of failure at the router and switch.
- Ping tests were run from PC0 only and not repeated.

4.8 Project Planning, Roles & Budget

Table 4.11 lists the project milestones. No calendar dates are given because none were recorded.

Table 4.11: Project milestones

Milestone	Description	Status
M1	Define requirements, scope and objectives	Completed
M2	Design the topology and addressing plan; compare alternatives	Completed
M3	Build the topology in Packet Tracer and check links	Completed
M4	Configure Router0 and static host settings	Completed
M5	Connectivity testing with ping	Completed
M6	Configure DHCP on Server0 and troubleshoot	Attempted; gateway assignment not verified
M7	Prepare the capstone report	Completed

Table 4.12: Roles and contribution

Student	Assigned Responsibility	Actual Contribution
RAMYA R (192312213)	All project tasks (sole student)	Design, simulation build, router and host configuration, testing, DHCP troubleshooting and report writing

Because this is an individual project, team coordination and leadership were not applicable beyond planning and managing the work independently.

Table 4.13: Budget for the simulated implementation

Item	Estimated Cost	Actual Cost
Cisco Packet Tracer simulation software	No purchase required	No cost recorded
Router, switch, four PCs and server (simulated devices)	No hardware purchase required	No hardware purchased
Ethernet cabling (simulated)	No hardware purchase required	No hardware purchased
Total	**No hardware purchase required**	**No hardware purchased**

A physical deployment would require real equipment. Table 4.14 shows the items that would be needed; no quotations were obtained, so no prices are given.

Table 4.14: Items required for a physical deployment (cost not estimated)

Item	Quantity	Cost Status
Cisco 1941 router	1	Not estimated; no quotation obtained
Cisco 2960-24TT switch	1	Not estimated; no quotation obtained
Desktop computers	4	Not estimated; no quotation obtained
Server	1	Not estimated; no quotation obtained
Copper Ethernet cables	6	Not estimated; no quotation obtained

CHAPTER 5
CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This project addressed the design and verification of a small office LAN in simulation. A star topology was designed and built in Cisco Packet Tracer with a Cisco 1941 router (Router0), a Cisco 2960-24TT switch (Switch0), four PCs (PC0–PC3) and one server (Server0), all in the subnet 192.168.1.0/24 with 192.168.1.1 as the default gateway. Router0 was configured on GigabitEthernet0/0, and Server0 and the PCs were given static IPv4 settings.

Testing showed green link indicators on the Ethernet connections, a successful ping from PC0 to Router0 with four replies and 0% packet loss, and successful replies reported from PC0 to PC1, PC2 and PC3. Router connectivity is therefore classified PASS and PC-to-PC connectivity PASS as reported. A DHCP pool named OfficePool was configured on Server0, but a client displayed gateway 0.0.0.0, so DHCP gateway assignment is NOT VERIFIED. Physical hardware performance was NOT TESTED. Objectives 1–5 and 7 were fully achieved and objective 6 was partially achieved.

The overall contribution is a documented, traceable baseline design that meets the core connectivity objectives in simulation and states its limitations openly.

5.2 Future Work

- Find and fix the DHCP fault, then verify that clients receive the gateway 192.168.1.1, DNS server 192.168.1.2 and an address from the pool, and check that leases do not collide with static addresses.
- Add DNS records on Server0 and test name resolution from the PCs.
- Test connectivity to Server0 and from the other PCs, and repeat the tests to build a fuller record.
- Add VLAN segmentation, access control lists, device passwords and secure remote management to address the security risks in Section 3.7.
- Add a second link or device to reduce single points of failure.
- Extend the network to an external connection and test internet access.
- Deploy the design on physical equipment and measure throughput and reliability.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- How a /24 IPv4 subnet is structured and how many hosts it supports.
- How a router interface becomes a default gateway (address, mask and no shutdown) and how write memory saves it.
- How DHCP pool parameters (gateway, DNS server, start address, maximum users) relate to the settings a client should receive.
- How to plan tests in advance and classify results as passed, not verified or not tested.

Engineering & Professional Skills Developed:

- Network design and addressing, command-line router configuration, and use of a network simulator.
- Systematic troubleshooting by isolating one change at a time.
- Honest technical reporting and clear documentation with figures, tables and references.

Individual Reflection – RAMYA R:

- **Most difficult problem.** The hardest problem I met was the DHCP service on Server0. A client showed a default gateway of 0.0.0.0 even though I had set the gateway 192.168.1.1 in the OfficePool settings. I did not find the cause, so I treated the DHCP gateway assignment as not verified and continued the tests with static addresses.
- **Key engineering decision.** I decided to use static addressing for the connectivity tests. The evidence was that PC0 pinged Router0 with four replies and 0% loss using static settings, which showed the router and switch part of the design was sound regardless of the DHCP problem.
- **New knowledge acquired independently.** I learned how the pool settings in a DHCP server correspond to what a client should receive, and that a missing gateway can stay hidden when only same-subnet pings are used.
- **What I would redesign.** If I repeated the project I would test DHCP step by step before building the rest, record the port used on each cable, and add tests for Server0 and DNS.

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APPENDICES

Appendix A – Detailed Calculations

Table A.1: Address calculations for 192.168.1.0/24

Quantity	Calculation	Result
Prefix length from mask 255.255.255.0	8 + 8 + 8 + 0 network bits	/24
Total addresses	2^(32 − 24) = 2^8	256
Network address	Host bits all 0	192.168.1.0
Broadcast address	Host bits all 1	192.168.1.255
Usable host addresses	256 − 2	254
Intended DHCP pool	Start .100, 50 users: .100 to .(100 + 50 − 1)	192.168.1.100 – 192.168.1.149
Static addresses in use	Router0 .1, Server0 .2, PC0–PC3 .10–.13	6 addresses
Unassigned and outside pool	254 − 6 − 50	198 addresses

Appendix B – Engineering Drawings / Schematics

The topology drawings are Figure 3.1 (architecture diagram), Figure 3.2 (address allocation) and Figure 4.1 (Packet Tracer screenshot). Table B.1 lists the connections between devices as drawn; an entry of “Yes” means a direct cable.

Table B.1: Direct device connections

Device	Router0	Switch0	PC0	PC1	PC2	PC3	Server0
Router0	—	Yes	No	No	No	No	No
Switch0	Yes	—	Yes	Yes	Yes	Yes	Yes
PC0 / PC1 / PC2 / PC3	No	Yes	No	No	No	No	No
Server0	No	Yes	No	No	No	No	—

Appendix C – Source Code / Code Repository Information

The project has no source code and no code repository. The configuration commands entered on Router0 are reproduced below.

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
end
write memory
```

Appendix D – Datasheets

The following manufacturer documents were consulted for device descriptions: the Cisco 1941 Series Integrated Services Routers Data Sheet [9] and the Cisco Catalyst 2960 Series Switches product bulletin [10]. No datasheet values are used as design inputs in this report.

Appendix E – Experimental / Raw Data

Table E.1 lists the implementation details and observations as recorded, without interpretation.

Table E.1: Recorded implementation observations

No.	Observation
1	Router0 was configured with IP address 192.168.1.1 and subnet mask 255.255.255.0.
2	The Ethernet connections showed green link indicators in Cisco Packet Tracer.
3	PC0 successfully pinged Router0 at 192.168.1.1, receiving four replies with 0% packet loss.
4	Successful replies were also reported when testing connectivity from PC0 to PC1, PC2 and PC3.
5	Server0 was assigned IP address 192.168.1.2, subnet mask 255.255.255.0 and default gateway 192.168.1.1.
6	DHCP configuration was attempted on Server0 using a pool named OfficePool.
7	The intended DHCP configuration used default gateway 192.168.1.1, DNS server 192.168.1.2, start IP address 192.168.1.100, subnet mask 255.255.255.0 and a maximum of 50 users.
8	During DHCP troubleshooting, a client displayed default gateway 0.0.0.0; DHCP gateway assignment was therefore not verified as working correctly.
9	Static IPv4 addressing was used for the successful connectivity tests.
10	The project was implemented in simulation, not deployed on a physical network.

Appendix F – Ethics / Institutional Approval

Not applicable. The project is a software simulation that involves no human participants, no personal data and no institutional approval requirement.

Appendix G – Project Meeting / Progress Records

No project meeting or progress records were available when this report was prepared, and none are included.

Appendix H – User/Industry Feedback

Not applicable. No external user or industry feedback was collected for this simulated project.

Appendix I – Publications / Patents / Competitions / Product Outcomes

None. The project has not resulted in any publication, patent, competition entry or product.

Appendix J – Individual Contribution Evidence

The project was completed by one student, RAMYA R (192312213). The evidence of individual work is the Packet Tracer topology in Figure 4.1, the configuration in Section 4.2 and Appendix C, and the test records in Tables 4.8 and E.1.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)